27 April 2026
BrightLearn Exercise 1
SQL fundamentals
SELECT & filtering

01 Selecting and Sorting.

1 Retrieve all columns from employees table

```
SELECT *
FROM employees
```

id	first_name	last_name	department	salary
1	Alice	Green	IT	70 000
2	Brian	Lee	HR	45 000
3	Cathy	Zulu	Finance	65 000
4	David	Mokoena	Marketing	50 000
5	Eva	Naidoo	IT	72 000

hire_date	city
2020 - 01 - 10	Johannesburg
2019 - 03 - 22	Cape Town
2018 - 07 - 18	Durban
2021 - 11 - 05	Pretoria
2017 - 09 - 30	Johannesburg.

02 Find all unique departments

```
SELECT DISTINCT department
FROM employees;
```

department
IT
HR
Finance
Marketing

03. Retrieve first and last names ordered by salary descending

```
SELECT first_name,
       last_name
FROM employees
ORDER BY salary DESC;
```

first_name	last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

4 Retrieve the top 3 highest paid employees

```
SELECT id,
       first_name,
       last_name,
       salary
```

```
FROM employees
ORDER BY salary DESC
LIMIT 3;
```

id	first_name	last-name	salary
5	Eva	Maidoo	72 000
1	Alice	Green	70 000
3	Cathy	Zulu	65 000

02 Filtering with WHERE

05 find employees in the IT department

```
SELECT    id,
          first-name,
          last-name,
          department
FROM employees
WHERE department = 'IT';
```

id	first-name	last-name	department
1	Alice	Green	IT
5	Eva	Maidoo	IT

06 Find employees in Finance with salary > 60 000

```
SELECT id,
       first_name,
       last_name,
       department, salary
FROM employees
WHERE department = 'Finance' AND salary > 60 000;
```

id	first_name	last_name	department	salary
3	Cathy	Zulu	Finance	65 000

07 Find employees in HR or Marketing

```
SELECT id,
       first_name,
       last_name,
       department
FROM employees
WHERE department = 'HR' OR department = 'Marketing';
```

id	first_name	last_name	department
2	Brian	Lee	HR
4	David	Molwena	Marketing

08 Find employees not in IT

```
SELECT id,
       first_name,
       last_name,
       department
FROM employees
WHERE department != 'IT';
```

id	first_name	last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Molobena	Marketing

09 Find employees in IT, HR, or Finance using IN

```
SELECT id,
       first_name,
       last_name,
       department
FROM employees
WHERE department IN ('IT', 'HR', 'Finance');
```

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Maidoo	IT

10 Find employees in IT with salary > 65 000 and city Johannesburg

```
SELECT id,
       first-name,
       last-name,
       department,
       salary,
       city
FROM employees
WHERE department = 'IT' AND salary > 65 000
      AND city = 'Johannesburg';
```

id	first-name	last-name	department	salary	city
1	Alice	Green	IT	70 000	Johannesburg
5	Eva	Naidoo	IT	72 000	Johannesburg